FIITJEE

ALL INDIA TEST SERIES

FULL TEST – IX

JEE (Main)-2024

TEST DATE: 27-03-2024

Time Allotted: 3 Hours **Maximum Marks: 300**

General Instructions:

- The test consists of total 90 questions.
- Each subject (PCM) has 30 questions.
- This question paper contains **Three Parts.**
- **Part-A** is Physics, **Part-B** is Chemistry and **Part-C** is Mathematics.
- Each part has only two sections: **Section-A and Section-B**.
- **Section – A** : Attempt all questions.
- **Section – B :** Do any five questions out of 10 questions.

Section-A (01 – 20, 31 – 50, 61 – 80) *contains 60 multiple choice questions which have* ***only one correct answer****. Each question carries* ***+4 marks*** *for correct answer and* ***–1 mark*** *for wrong answer.*

Section-B (21 – 30, 51 – 60, 81 – 90) *contains 30 Numerical based questions. The answer to each question is rounded off to the nearest integer value. Each question carries* ***+4 marks*** *for correct answer and* ***–1 mark*** *for wrong answer.*

Physics

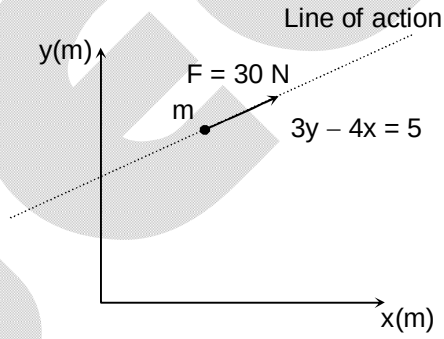
PART – A

SECTION – A
(One Options Correct Type)

This section contains **20 multiple choice questions**. Each question has **four choices** (A), (B), (C) and (D), out of which **ONLY ONE** option is correct.

1. An object of density d_0 kept deep inside water of density d_w and released. During the time it moves a vertical distance h within water.
(A) The gravitational P.E. of water in the vessel increases if $d_0 < d_w$
(B) The gravitaitonal P.E. of the object increases if $d_0 > d_w$
(C) The gravitational P.E. of the object decreases if $d_0 < d_w$
(D) If entire system is kept on weighing machine, then the reading of weighing machine is less than weight of the system after releasing the ball (in each case either $d_0 > d_w$ or $d_0 < d_w$).

2. A force of magnitude 30 N acting along the line $3y - 4x = 5$ in x-y plane as shown. There are two points P(1, 2) and Q(1, 3)



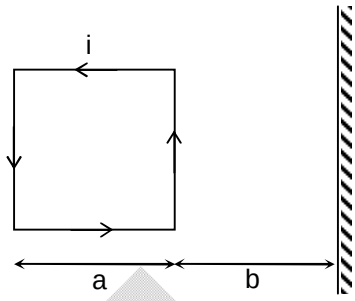
(A) $\vec{\tau}_P = 18\ (N-m)(-\hat{k}),\ \vec{\tau}_Q = \vec{0}$
(B) $\vec{\tau}_P = 18\ (N-m)(\hat{k}),\ \vec{\tau}_Q = \vec{0}$
(C) $\vec{\tau}_P = \vec{0},\ \vec{\tau}_Q = 18\ (N-m)(-\hat{k})$
(D) $\vec{\tau}_P = \vec{0},\ \vec{\tau}_Q = 18\ (N-m)(\hat{k})$

3. A particle has acceleration $\vec{a} = 6\hat{i} + \alpha\hat{k}$ (where α is constant) due to a magnetic field $\vec{B} = -3\hat{i} + \hat{j} + 2\hat{k}$. Find the value of α.
(A) 18 (B) –8
(C) 9 (D) –9

4. In a uniform electric field and magnetic field, net force on a moving charge is zero, then
(i) $\vec{E}$ must be perpendicular to $\vec{B}$
(ii) $\vec{E}$ must be perpendicular to $\vec{v}$
(iii) $\vec{B}$ must be perpendicular to $\vec{v}$
(iv) $\vec{E}$, $\vec{v}$ & $\vec{B}$ must be perpendicular to each other
(A) (i), (ii) are correct (B) (i), (ii), (iii) are correct
(C) (i), (ii), (iv) are correct (D) (i), (ii), (iii) and (iv) are correct

5. A light planet is revolving around a massive star in a circular orbit of radius R with a period of revolution T. If the force of attraction between planet and star is proportional to $R^{-3/2}$ then choose the correct option:
(A) $T^2 \alpha R^{7/2}$ (B) $T^2 \alpha R^3$
(C) $T^2 \alpha R^{5/2}$ (D) $T^2 \alpha R^{3/2}$

6. A square coil of side 'a' carrying current 'i' placed at a distance 'b' from a long wall. Find the magnetic flux that passes through a region which is located to the right of the wall.



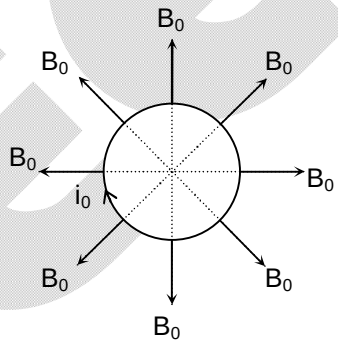
(A) $\frac{\mu_0 ib}{2\pi}\ell n\left(\frac{a+b}{b}\right)$

(B) $\frac{\mu_0 ia}{2\pi}\ell n\left(\frac{a+b}{a}\right)$

(C) $\frac{\mu_0 ib}{4\pi}\ell n\left(\frac{a+b}{b}\right)$

(D) $\frac{\mu_0 ia}{4\pi}\ell n\left(\frac{a+b}{a}\right)$

7. Consider a hypothetical radial magnetic field as shown in the figure. Find net magnetic force on the closed current carrying ring (i_0 in cw sense) (given : B_0 at the site of current carrying ring)



(A) zero

(B) $i(2\pi R)B_0 \otimes$

(C) $i(2\pi R)B_0 \odot$

(D) can not be calculated

8. A particle having mass 'm' and charge 'Q' is placed in a uniform magnetic field $B_0(-\hat{k})$, it breaks into two fragments, in its state of rest. The two fragments move in circles of radii 'R' and '2R' if λ_1 and λ_2 are their de-Broglie wavelength and q_1 and q_2 are charge on two fragments. Then choose the correct option.

(A) $\frac{\lambda_1}{\lambda_2}=1,\ \frac{q_1}{q_2}=2$

(B) $\frac{\lambda_1}{\lambda_2}=2,\ \frac{q_1}{q_2}=1$

(C) $\frac{\lambda_1}{\lambda_2}=2,\ \frac{q_1}{q_2}=1$

(D) $\frac{\lambda_1}{\lambda_2}=\frac{1}{2},\ \frac{q_1}{q_2}=2$

9. A hydrogen like atom is in excited state of principal quantum number 'n' and come to first excited state by successive emitting two photons of energy 10.2 eV and 17 eV. This atom from the same excited state (i.e. principal quantum number 'n') can make a transition to second excited state by successively emitting two photons of energy 4.25 eV and 5.95 eV respectively. Then choose the correct option.

(A) n = 3, z = 6

(B) n = 6, z = 3

(C) n = 4, z = 6

(D) n = 6, z = 4

10. In a spherically symmetric region of radius R, electric field varies as $E = Ar^2$ $(0 \le r \le R)$ where r is radial distance. Find the variation of density of charge.

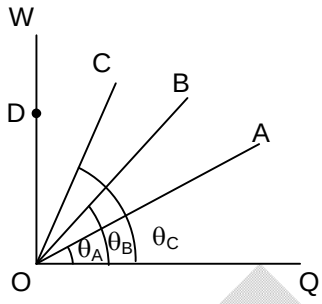
(A) ρ = constant

(B) $\rho = 4\varepsilon_0 Ar$

(C) $\rho = \varepsilon_0 Ar$

(D) $\rho = \varepsilon_0 Ar^2$

11. Figure shows four lines OA, OB, OC, OD (representing various processes such as isothermal, adiabatic, isobaric for a monoatomic and diatomic ideal gas) on a work done by the gas (W) versus heat supplied (Q) graph. The initial states of both the gases are the same and also the scale for the two axes. Then choose the **INCORRECT** statement.



$\tan\theta_A = \frac{2}{7}$

$\tan\theta_B = \frac{2}{5}$

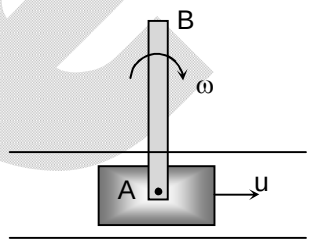
$\tan\theta_C = 1$

(A) OA represents isobaric process for diatomic gas
(B) OB represents isobaric process for diatomic gas
(C) OC represents isothermal process for monoatomic gas
(D) OD represents adiabatic process for diatomic gas

12. A force exerts an impulse I on a particle changing its speed from U to 2U. The applied force and the velocity are oppositely oriented along the same line. The work done by the force is

(A) $\frac{3}{2}$IU (B) $\frac{1}{2}$IU
(C) IU (D) 2IU

13. A mechanism consists of a part which is translated with a velocity 'u' and a rod AB of length L and mass M hinged at A. The rod rotates about A with angular velocity ω. The kinetic energy of the rod (with respect to ground frame) when it is vertical as shown is:

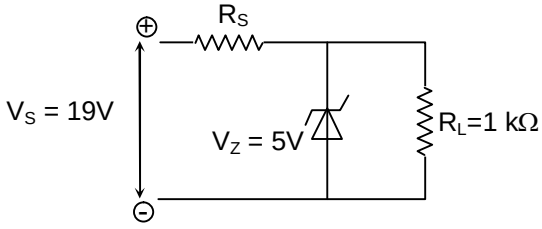


(A) $\frac{1}{2}Mu^2 + \frac{1}{6}ML^2\omega^2$
(B) $\frac{1}{2}Mu^2 + \frac{1}{6}ML\omega u$
(C) $\frac{1}{2}Mu^2 + \frac{1}{6}ML^2\omega^2 + \frac{1}{2}ML\omega u$
(D) None of these

14. A particle moving in a circle of radius R with uniform speed takes time T to complete one revolution. If this particle is projected at an angle θ to the horizotnal with the same speed, horizontal range attained by it is equal to 2R. The angle of projection 'θ' is given by

(A) $\frac{1}{2}\sin^{-1}\left(\frac{2gT^2}{\pi^2R}\right)$ (B) $\frac{1}{2}\sin^{-1}\left(\frac{gT^2}{2\pi^2R}\right)$
(C) $\frac{1}{2}\cos^{-1}\left(\frac{2gT^2}{\pi^2R}\right)$ (D) $\frac{1}{2}\cos^{-1}\left(\frac{gT^2}{2\pi^2R}\right)$

15. In the given circuit if the power rating of Zener diode is 10 mW, the value of series resistance R_s to regulate the input unregulated supply is



(A) 500 Ω
(B) 1000 Ω
(C) 1500 Ω
(D) 2000 Ω

16. In an electron gun, emission current from the cathode is given by the equation

$I = AT^2 e^{-\left(\frac{B}{KT}\right)}$

(K = Boltzmann constant, A = constant)
The dimension formula for AB^2 is same as

(A) [kT] (B) $[IT^2]$

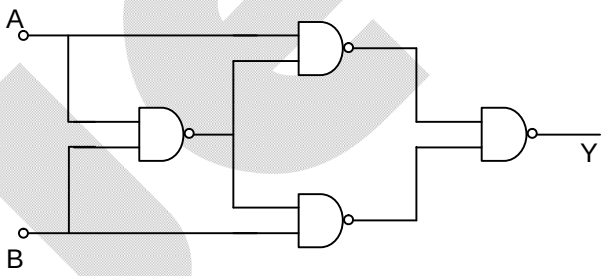
(C) $[IK^2]$ (D) $\left[\frac{IK^2}{T}\right]$

17. The electric field of a plane electromagnetic wave is given by $\vec{E} = E_0\hat{i}\cos(kz)\cos\omega t$. The corresponding magnetic field B is given by

(A) $\vec{B} = \frac{E_0}{C}\hat{j}\sin(kz)\cos\omega t$ (B) $\vec{B} = \frac{E_0}{C}\hat{j}\sin(kz)\sin\omega t$

(C) $\vec{B} = \frac{E_0}{C}\hat{k}\sin(kz)\cos\omega t$ (D) $\vec{B} = \frac{E_0}{C}\hat{j}\cos(kz)\sin\omega t$

18. The output of the given circuit diagram is



(A)

A	B	Y
0	0	0
0	1	0
1	0	1
1	1	0

(B)

A	B	Y
0	0	0
0	1	1
1	0	0
1	1	1

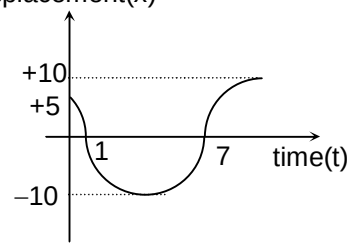
(C)

A	B	Y
0	0	0
0	1	1
1	0	1
1	1	0

(D)

A	B	Y
0	0	1
0	1	1
1	0	1
1	1	0

19. Displacement time graph of particle executing SHM is shown in the figure. The equation of SHM is given by



(A) $x = 10\sin\left(\frac{\pi t}{2} + \frac{5\pi}{6}\right)$

(B) $x = 10\sin\left(\frac{\pi t}{6} + \frac{5\pi}{6}\right)$

(C) $x = 10\sin\left(\frac{\pi t}{6} + \frac{\pi}{6}\right)$

(D) $x = 10\sin\left(\frac{\pi t}{3} + \frac{\pi}{6}\right)$

20. There is a concave mirror of focal length 'f'. A ray is incident parallel to the principal axis and at a distance 'f' from principal axis. Find out the point where it meets the principal axis from pole.

(A) $2f\left(2 - \frac{1}{\sqrt{3}}\right)$ (B) $2f\left(1 - \frac{1}{\sqrt{3}}\right)$

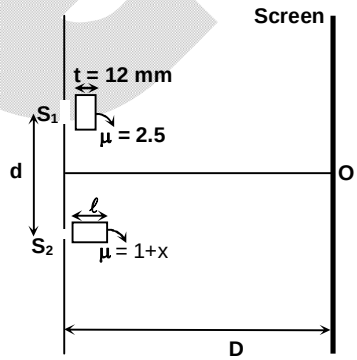
(C) $f\left(2 - \frac{1}{\sqrt{3}}\right)$ (D) $f\left(1 - \frac{1}{\sqrt{3}}\right)$

SECTION – B

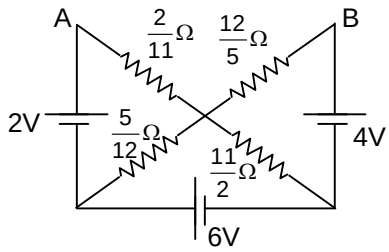
(Numerical Answer Type)

This section contains **10** Numerical based questions. The answer to each question is rounded off to the nearest integer value.

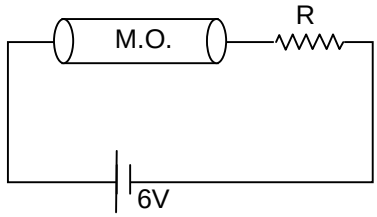
21. In young's experiment two slabs are placed in front of the pinholes as shown in the figure. Refractive index of second slab is varies as $\mu = 1 + x$, where 'x' is distance from left end of the slab. Find the length of the slab (in mm) so that central maxima gets formed at 'O'. (ℓ,d, t $\ll$ D)



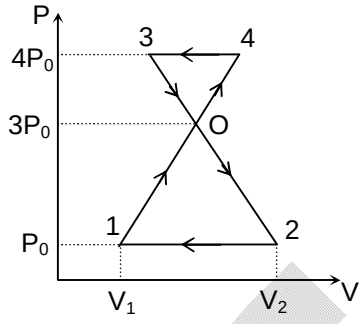
22. In the circuit shown, the value of $V_A - V_B$ = ...V



23. A rod of metallic oxide is connected in series with a resistance R and both are connected to a battery of emf 6V as shown in the figure. The potential difference across the rod of metallic oxide depends on current flowing to it according to the relation $\sqrt{V} = i$. If power dissipated in rod of metallic oxide is double the power dissipated in resistance. The value of R is .....Ω



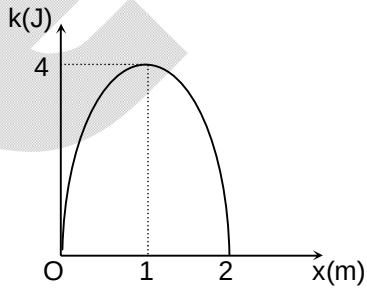
24. In the figure shown, work done by gas is …….J. (given $P_0 = 10^5$ Pa and $V_2 - V_1 = 10$ Lt)



25. The energy of a system as a function of time 't' is given as $E(t) = A^2e^{-\alpha t}$, where $\alpha = 0.2\ s^{-1}$. the measurement of A has an error of 1.25 %. If error in the measurement of time is 1.5%, then percentage error in the value of E(t) at t = 5 sec is

26. A string fixed at both ends vibrates in a resonance mode with a separation of 2 cm between two consecutive nodes, for the next higher resonance frequency this separation is reduced to 1.6 cm. Then length of string (in cm).

27. A block of mass m lies on a horizontal frictionless surface and is attached to one end of a horizontal spring (with spring constant K) whose other end is fixed. The block is initially at rest at the position where the spring is unstretched (x = 0). A constant force $\vec{F}$ in the positive direction of the x-axis is applied to it, a plot of the resulting kinetic energy of the block versus its position x is shown in the figure. The magnitude of $\vec{F}$ is……..N.



28. Net power supplied to a particle of mass 2 kg is $P = 1.5\ t^2$. If at t = 0, particle is at rest. Then its speed at t = 2 sec is ……..(m/s)

29. The distance between object and its 3 times magnified virtual image as produced by a convex lens is 20 cm. The focal length of the lens used is …….(cm).

30. A 2μF capacitor is charged by a 10 V battery and 3 μF capacitor is charged by 30 V battery,. Now battery is removed and capacitor plates of opposite polarity are connected to form a closed loop. Heat loss after switched is closed is ……..(mJ).

Chemistry

PART – B

SECTION – A
(One Options Correct Type)

This section contains **20 multiple choice questions**. Each question has **four choices** (A), (B), (C) and (D), out of which **ONLY ONE** option is correct.

31. Ratio of radius of 3rd Bohr orbit of Li^{2+} to 4th Bohr radius of C^{+5} is

(A) $\frac{3}{1}$ (B) $\frac{3}{2}$

(C) $\frac{9}{8}$ (D) $\frac{3}{4}$

32. Which of the following compound is diamagnetic?

(A) $\left[Mn(CN)_6\right]^{3-}$ (B) $\left[Cr(H_2O)_6\right]^{3+}$

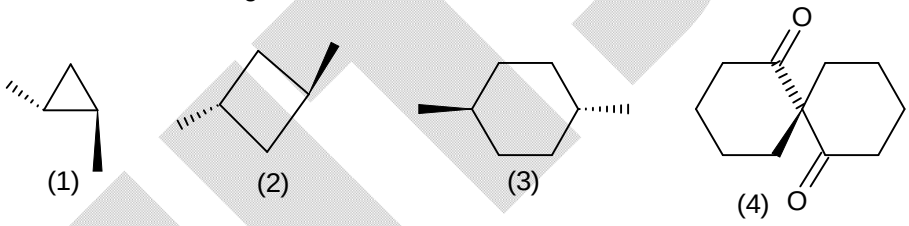
(C) $\left[Co(NH_3)_6\right]^{3+}$ (D) $[FeF_6]^{4-}$

33. The equilibrium constant for the reaction $A \rightleftharpoons B$ is 2×10^3. The forward rate constant in the presence and absence of catalyst was found to be 5×10^4 sec^{-1} and 4×10^{-6} sec^{-1} respectively. The expected rate constant for the reverse reaction in the absence of catalyst is:

(A) 5×10^{10} (B) 8×10^{-3}

(C) 2×10^{-9} (D) 4×10^{-2}

34. Which of the following molecules are chiral?



(A) 1 and 3 (B) 1 and 4

(C) 2 and 3 (D) 2, 3 and 4

35. The bond order in carbon monoxide (CO) decreases when bonded to transition metal in their low oxidation state because

(A) $\pi-$electrons of CO are donated to central metal.

(B) $\sigma-$electrons of CO are donated to central metal.

(C) d-electrons of central metal are transferred to empty π^* orbitals of CO.

(D) p-electrons of central metal are transferred to empty π orbitals of CO.

36. **Assertion (A):** The tendency to show catenation decreases down the group in group-14 elements.

Reason (R): Down the group size increases and electronegatively decreases.

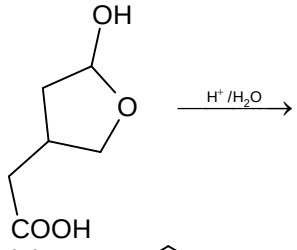
(A) (A) is true (R) is false.

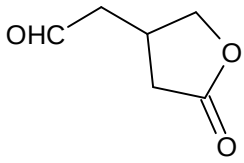
(B) (A) is false (R) is true.

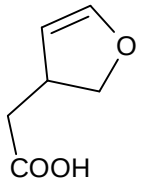
(C) Both (A) and (R) are true and (R) is the correct explanation of (A).

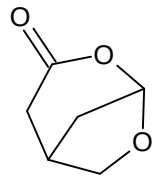
(D) Both (A) and (R) are true and (R) is not the correct explanation of (A).

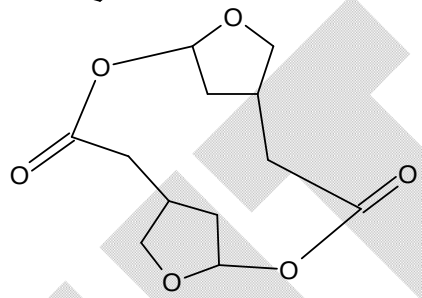
37. The major product formed in the following reaction is



(A) 

(B) 

(C) 

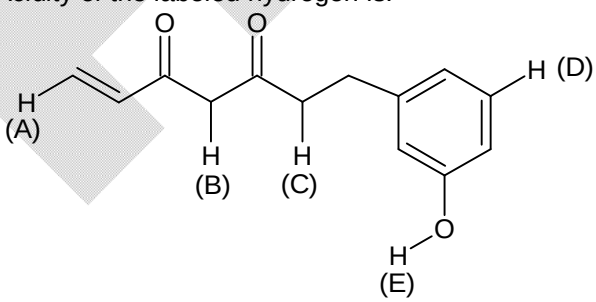
(D) 

38. Radium $\left({}_{88}Ra^{224}\right)$ decays by a series of emission of three $\beta-$particles and two $\alpha-$particles. The end product 'X' is

(A) ${}_{88}X^{220}$ (B) ${}_{87}X^{216}$

(C) ${}_{90}X^{234}$ (D) ${}_{88}X^{216}$

39. Acidity of the labeled hydrogen is:



(A) B > E > C > A > D (B) B > E > C > D > A

(C) E > A > B > C > D (D) E > B > C > A > D

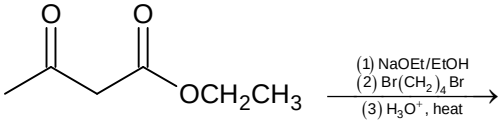
40. Which of the following reaction doesn't give H_3PO_4 ?

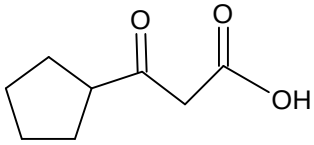
(A) $P_4O_6 + H_2O \longrightarrow$ (B) $P_4S_{10} + H_2O \longrightarrow$

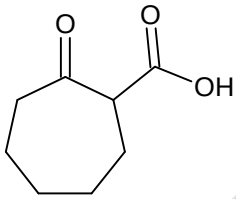
(C) $P_2O_5 + H_2SO_4 \longrightarrow$ (D) $PCl_5 + H_2O \longrightarrow$

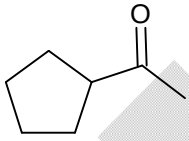
41. **Assertion (A):** Borax bead test is employed only for coloured salts.
Reason (R): Borax reacts with metal salts to form metal borates or metals, which have characteristics colours.
(A) (A) is true (R) is false.
(B) (A) is false (R) is true.
(C) Both (A) and (R) are true and (R) is the correct explanation of (A).
(D) Both (A) and (R) are true and (R) is not the correct explanation of (A).

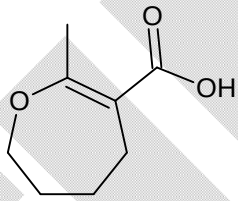
42. The major product in the following reaction is



(A)


(B)


(C)


(D)


43. The complex that exist as a pair of enantiomers is

(A) $\text{cis} - \left[\text{Co}\left(\text{NH}_3\right)_4\text{Cl}_2\right]^{4+}$ (B) $\left[\text{Pt}\left(\text{PPh}_3\right)\left(\text{Cl}\right)\left(\text{Br}\right)\left(\text{CH}_3\right)\right]^-$

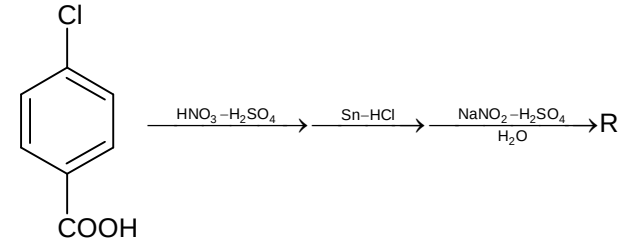
(C) $\text{trans} - \left[\text{Co}\left(\text{H}_2\text{NCH}_2\text{CH}_2\text{NH}_2\right)_2\text{Cl}_2\right]^+$ (D) $\left[\text{Co}\left(\text{H}_2\text{NCH}_2\text{CH}_2\text{NH}_2\right)_3\right]^{3+}$

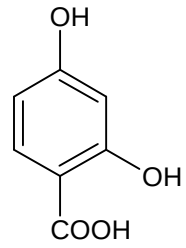
44. A large hot water bath at 360 K looses 60 Joules of heat energy reversibly to the surrounding which is at 300 K. The total entropy change $\left(\text{in JK}^{-1}\right)$ would be

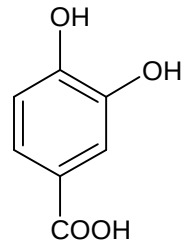
(A) + 0.33 (B) –0.033

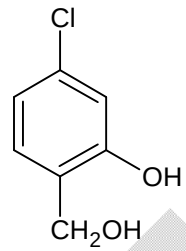
(C) + 0.20 (D) + 0.033

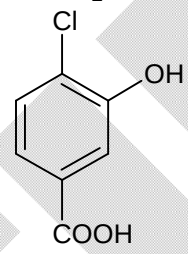
45. The major product 'P' formed in the following reaction sequence is



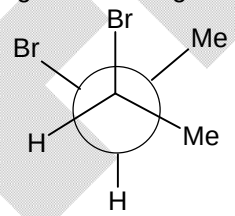
(A) 

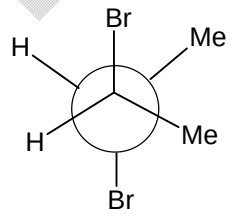
(B) 

(C) 

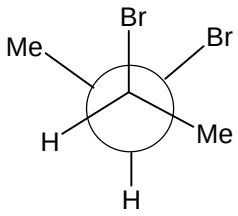
(D) 

46. Among the following the most stable conformation of meso-2, 3 dibromobutane is

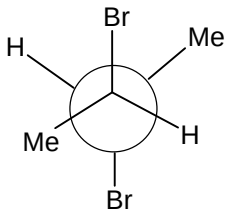
(A) 

(B) 

(C)



(D)



47. Filter paper moistioned with ammonical sodium nitroprusside solution turns violet on contact with a drop of alkaline Na_2S solution. The violet colour is due to the formation of

(A) $\left[Fe(SCN)_5(NO)\right]^-$ (B) $\left[Fe(SCN)_5(NO)\right]^{2-}$

(C) $\left[Fe(CN)_5(NOS)\right]^{3-}$ (D) $\left[Fe(CN)_5(NOS)\right]^{4-}$

48. The pK_a of a monobasic organic acid is 4.2. The pH at which 80% of the acid will be in ionized formed is

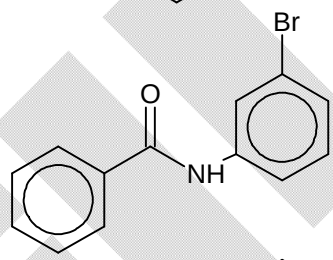
(A) 4.8 (B) 4.1

(C) 3.4 (D) 2.9

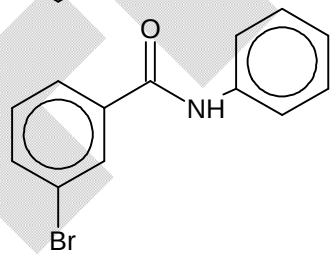
49. Major product of the following reaction is



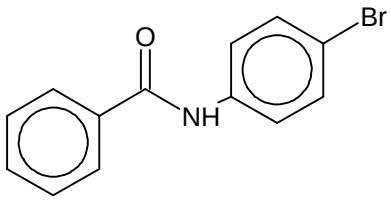
(A)



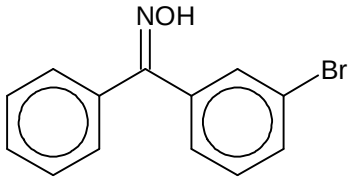
(B)



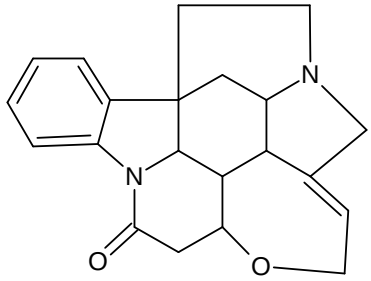
(C)



(D)



50. Number of chiral carbons in a molecule of strychnine are_____



strychnine

(A) 6 (B) 4
(C) 5 (D) 8

SECTION – B

(Numerical Answer Type)

This section contains **10** Numerical based questions. The answer to each question is rounded off to the nearest integer value.

51. 6 moles of compound A are reacted with excess of B in the following reaction sequence

$2A + B \longrightarrow C$

$3C \longrightarrow 2D$

34 g of D is obtained at the end of reaction. The molar mass of D is ……gm/mol.

52. For $\left[Ti(H_2O)_6\right]^{3+}$, the absorption maximum due to d – d transition is found at 20,000 cm^{-1}. The magnitude of crystal field stabilization energy is ____$\times 10^3$ cm^{-1}.

53. The oxidation state of sulphur in dithionous acid is_____

54. A tranquilizer 'T' is synthesized according to the following reaction scheme. The molar mass of the tranquilizer T is_____gm/mol (H = 1, C = 12, N = 14, O = 16, Cl = 35.5)

$$C_8H_{16} \xrightarrow[Zn/H_2O]{O_3} 2Q \xrightarrow[\Delta]{NaOH} R \xrightarrow{Ag_2O/NaOH} S \xrightarrow[(iii)\,CF_3COOH]{(i)\,SOCl_2 \;(ii)\,NH_3} T$$

55. The molar conductivities at infinite dilution Λ_m^o for $Na_2SO_4, K_2SO_4, KCl, HCl$ and HCOONa at 300 K are 260, 308, 50, 426 and 105 $Scm^2\,mol^{-1}$ respectively. The Λ_m^o for formic acid is _____ $Scm^2\,mol^{-1}$.

56. The molar mass of lowest alkane(having lowest molar mass) which gives four structural isomers on monochlorination is_____gm/mol.

57. If 0.01 M solution of a non-electrolyte 'X' provides an osmotic pressure of 'P' atm then 0.1 M solution of strong binary electrolyte 'Y' at the same temperature will produce an osmotic pressure of _____× P atm.

58. The enthalpy of formation of OH^- ions at 25°C from the following thermochemical data is _____ kJ (in magnitude)

$H_2O(\ell) \longrightarrow H^+(aq) + OH^-(aq)$ $\quad \Delta H^o = 57.3$ kJ

$H_2(g) + \frac{1}{2}O_2(g) \longrightarrow H_2O(\ell)$ $\quad \Delta H^o = -285.3$ kJ

and $\Delta H_f^o(H^+) = 0.0$ kJ

59. For a homogeneous gaseous reaction $2XY_2 \rightleftharpoons 2XY + Y_2$ the value of equilibrium constant K_P is 1.6×10^{-5} atm and the total pressure at equilibrium is 4 atm. The degree of dissociation 'α' of XY_2 is (assuming $\alpha << 1$)_____%.

60. Maximum CFSE value for tetrahedral complexes is xDq. The value of |x| is

Mathematics

PART – C

SECTION – A
(One Options Correct Type)

This section contains **20 multiple choice questions**. Each question has **four choices** (A), (B), (C) and (D), out of which **ONLY ONE** option is correct.

61. Let x, y, z > 1 and $A = \begin{bmatrix} 1 & \log_x y & \log_x z \\ \log_y x & 3 & \log_y z \\ \log_z x & \log_z y & 4 \end{bmatrix}$, if |adj(adj A)| = k, then k is divisor of

(A) $\frac{(3!)!}{(4!)^{3!}}$ (B) $\frac{(5!)!}{(4!)^{5!}}$

(C) $\frac{(4!)!}{(4!)^{3!}}$ (D) $\frac{(4!)!}{(3!)^{4!}}$

62. If $(f(x) - 1)(x^4 + x^2 + 1)^2 - (f(x) + 1)(x^8 + x^4 + 1) = 0 \ \forall\ x \in R - \{0\}$ and $f(x) \neq \pm 1$, then which of the following statement is incorrect ?

(A) $f(x) \geq 2 \ \forall\ x \in R - \{0\}$ (B) f(x) has a local minimum at x = –1

(C) f(x) has local maximum at x = 2 (D) $\int_{-1}^{1} x^2 f(x)\,dx = \frac{12}{5}$

63. If $L = \left|\int_0^{\pi} \frac{x^2 \tan x \sec^2 x\,dx}{(1+\tan^4 x)}\right|$, then the value of [L] is (where [.] is greatest integer function)

(A) 5 (B) 4

(C) 8 (D) 3

64. If $\int \frac{\sec\theta\,d\theta}{\tan^3\theta(1+\sec^2\theta)^2} = \frac{1}{4\sec^3\theta(1-\sec^4\theta)} - \frac{3}{4}F(\theta)$ and $F\left(\frac{\pi}{3}\right) = -\frac{1}{4}\ln\alpha - \frac{1}{2}\tan^{-1}\beta + \frac{1}{\gamma}$, then the value of $\alpha + \beta + \gamma$ =

(A) 14 (B) 29

(C) 19 (D) 0

65. An ellipse and a hyperbola intersect at four points orthogonally. The ellipse has vertices at the origin and (10, 24) and a semi-minor axis of length 11. The hyperbola has a conjugate axis of length $4\sqrt{3}$. If F_1, F_2 are the foci of the ellipse and P is one of the point of intersection. Then the value of $(PF_1)^2 + (PF_2)^2$ is

(A) 432 (B) 406

(C) 396 (D) 410

66. The term independent of x in the expansion of $\left(x^2 + x^4 + \frac{1}{x^2} + \frac{1}{x^4}\right)^{15}$ is

(A) $\sum_{r=0}^{5} {}^{15}C_{5+r}\,{}^{15}C_{5-r}$ (B) ${}^{30}C_{15}$

(C) $\sum_{r=0}^{5} {}^{15}C_{5-r}\,{}^{15}C_{3r}$ (D) $\sum_{r=0}^{5} {}^{15}C_{5+r}\,{}^{15}C_{3r}$

67. A circle of radius 3 units is inscribed in a parallelogram ABCD such that a smaller circle touches it externally and also touches the two sides of the parallelogram. The distance between the points of tangency which lie on the same side of ABCD is equal to 3 units. If d_1 and d_2 are the diagonals of ABCD, then the value of $(d_1 + d_2)^2$ is equal to

(A) 775 (B) 425
(C) 645 (D) 825

68. Let the solution curve y = f(x) of the differential equation $\frac{dy}{dx}+\frac{xy}{x^2-1}=\frac{15x^4+4x^3+4x+1}{\sqrt{1-x^2}}$, $x \in (-1, 1)$ passes through the origin. Then $\int_{-1/2}^{1/2}\frac{f(x)dx}{(x^2+2)}$ is equal to

(A) $\frac{\pi}{6}$ (B) $\frac{\pi}{6}-\frac{1}{4}$
(C) $\frac{\pi}{3}-\frac{\sqrt{3}}{4}$ (D) $\frac{\pi}{3}-\frac{1}{4}$

69. If $L = \lim_{x\to\infty}\left[(x+\cos A)(x+\cos B)(x+\cos C)\right]^{1/3} - x$, where A, B, C are the vertices of a ΔABC. Then L is equal to

(A) $\frac{\cos A\cos B\cos C}{3}$ (B) $\frac{4}{3}\sin\frac{A}{2}\sin\frac{B}{2}\sin\frac{C}{2}$
(C) $\frac{4}{3}\cos\frac{A}{2}\cos\frac{B}{2}\cos\frac{C}{2}$ (D) $\frac{1+4\sin\frac{A}{2}\sin\frac{B}{2}\sin\frac{C}{2}}{3}$

70. If m is selected at random from set {–4, –3, –2, –1, 0, 1, 2, 3, 4,} and the probability that the quadratic equation $x^2 + 2(m - 1)x + m + 5 = 0$ has one root greater than 3 and the other root smaller than 1 is k. Then the value of k is

(A) 1/4 (B) 1/3
(C) 2/3 (D) 7/9

71. Let p, q, r be non-zero real numbers such that they are 10th, 100th, 1000th terms of a H.P. Consider the system of linear equations :

$x + y + z = 1$
$10x + 100y + 1000z = 0$
$(qr)x + (pr)y + (pq)z = 0$, then

(A) if $\frac{q}{r}=10$, then system has no solution

(B) if $\frac{p}{r}\neq 100$, then system has infinitely many solution

(C) if $\frac{p}{q}\neq 10$, then system has atleast one solution

(D) $\frac{p}{q}=10$, then system has $x=\frac{10}{9}, y=-\frac{1}{9}, z=0$ as a solution

72. If f(x) is a differentiable function such that it satisfies $(x - y) f(x + y) - (x + y) f(x - y) = 2(x^2y - y^3)$ $\forall\ x, y \in R$, then which of the following is incorrect ?

(A) f(x) cuts x-axis at exactly two distinct points

(B) f(x) has a local maxima at $x = -\frac{1}{2}$

(C) area bounded by f(x) with x-axis is $\frac{2}{3}$ sq. units

(D) $\lim\limits_{x\to1} f(x) = 2$

73. The sum to infinite terms of the series :

$\tan^{-1}(2) + \tan^{-1}\frac{4}{1-2^2+2^4} + \tan^{-1}\frac{6}{1-3^2+3^4} + \ldots\ldots$ is

(A) $\frac{\pi}{4}$ (B) $\frac{\pi}{2}$

(C) $\frac{3\pi}{4}$ (D) none of these

74. If in a ΔABC, coordinates of its orthocentre and circumcentre are (9, 5) and (0, 0) respectively, and equation of side BC is 2x – y = 10. Then the value of $\int_a^b \frac{[x^2]}{[x^2-20x+100]+[x^2]}dx$ (where [.] is greater integer function and (a, b) is the coordinate of vertex A) is

(A) 4 (B) 0

(C) 8 (D) 1

75. Let the mean and variance of 6 observations a, b, 68, 44, 48, 60 be 55 and 194 respectively. If a > b, then a – b is

(A) 75 (B) 35

(C) 45 (D) 40

76. Let f : (0, π) $\to$ R be twice differentiable function such that $\lim\limits_{t\to x}\frac{f(x)\sin t - f(t)\sin x}{t-x} = \sin^2 x$ for $\forall$ x $\in$(0, π). If $f\left(\frac{\pi}{6}\right) = \frac{\pi}{12}$, then which is correct ?

(A) $f\left(\frac{\pi}{4}\right) = -\frac{\pi}{4\sqrt{2}}$ (B) $f\left(\frac{\pi}{4}\right) < \frac{x^4}{6} - x^2 \quad \forall\, x \in (0, \pi)$

(C) there exist $\alpha \in (0, \pi)$ such that f'(α) = 0 (D) $f''\left(\frac{\pi}{2}\right) - f\left(\frac{\pi}{2}\right) = \pi$

77. Tangents are drawn at the extremities of a variable chord ax + by + c = 0 (where a, b, c are in A.P.) to a parabola $y^2 = 4x$. The locus of the point of intersection of these tangents is

(A) x + 1 = 0 (B) a pair of straight line

(C) x + y + 1 = 0 (D) none of these

78. Area bounded by $y = \frac{2}{\pi}\left[|\cos^{-1}(\sin x)| - |\sin^{-1}(\cos x)|\right]$ and x-axis between $\frac{3\pi}{2} \le x \le 2\pi$ (where [.] greatest integer function) is

(A) $\frac{\pi}{4}$ (B) 2π

(C) $\frac{\pi}{2}$ (D) π

79. Let $\vec{a} = \hat{i} + 2\hat{j} + 3\hat{k}, \vec{b} = 4\hat{i} - 3\hat{j} + 2\hat{k}$ and $\vec{c}$ be the vector such that $\vec{a}\times\vec{c} = \vec{b}$ and $\vec{a}\cdot\vec{c} = 3$. Then $\vec{a}\cdot\left((\vec{c}\times\vec{b}) - \vec{b} - \vec{c}\right)$ is equal to

(A) 20 (B) 26

(C) 22 (D) 24

80. Let $C_1 \equiv (x-1)^2 + y^2 = 1$ and $C_2 : x^2 + y^2 = r^2$ intersect at P in 1st quadrant and let R be the point where the line passing through the points $Q \equiv (0, r)$ and P intersects the x-axis. If radius of C_2 decreases infinitely then limiting position of R is given by $(\alpha, 0)$ then 'α' is equal to
(A) 1 (B) 2
(C) 3 (D) 4

SECTION – B

(Numerical Answer Type)

This section contains **10** Numerical based questions. The answer to each question is rounded off to the nearest integer value.

81. From a point $P(\lambda, \lambda, \lambda)$ perpendicular PQ and PR are drawn respectively on the lines $\frac{x+7}{-6} = \frac{y-6}{7} = z$ and $\frac{7-x}{2} = y - 2 = z - 6$. If P is such that $\angle QPR$ is 90°, then the sum of possible values of λ is k, then value of 127k is

82. How many six lettered words (with or without meaning) can be formed using the letters of the word **ACCOMMODATION**

83. Let $A(z_1)$, $B(z_2)$, $C(z_3)$ be the vertices of a ΔABC such that $z_1 = 3 + 4i$, $z_2 = \frac{5}{13} + i\left(\frac{12}{13}\right)$, $z_3 = 8 + 15i$. It is given that $x_r = z_1 + z_2\omega^{r-1} + z_3\omega^{2r-2}$ where r = 1, 2, 3 and ω is a non-real cube root of unity then $\sum_{r=1}^{3} |x_r|^2 = k$. Find the value of k.

84. The total number of 3×3 matrices A having entries from the set {0, 1, 2, 3} such that $tr(AA^T) = 10$ is

85. Let $x_1, x_2, \ldots, x_9$ be the distinct roots of the polynomial $f(x) = x^9 - 9$. If $k = \prod_{1 \le i < j \le 9} (x_i + x_j)$, then find the value of $\frac{k^2}{9^5}$

86. The sum of the common terms of the following three AP's
1, 5, 9, …… , 404
4, 7, 10, …… , 304
4, 9, 14, …… , 149

87. If $(2025)^{3762}$ is divided by 17, then the remainder is

88. Let $A = \{x : x^2 \le x + 20000 ; x \in N\}$
$B = \{x = 3k + 2 ; k \in N\}$
$C = \{x = 4k ; k \in N\}$
The sum of the elements of the set $A \cap (B - C)$ is

89. The number of symmetric relations defined on the set {5, 6, 7 8, 9} which are not reflexive is

90. Vectors $\vec{a} = 2\lambda\hat{i} + 4\lambda\hat{j} + \hat{k}$ and $\vec{b} = 7\hat{i} - 2\hat{j} + \lambda\hat{k}$ make an acute angle whereas $\vec{b}$ make an angle θ with the positive z-axis which is less than $\frac{\pi}{6}$. Then the least positive integral value of λ will be